Sequencial Search And Program In Python

Essential Of Data Science

DEFINATION

Sequential search, also called linear search, is the simplest searching technique.

It checks each element in a list one by one until the required element is found or the list ends.

It works on both sorted and unsorted data.

HOW SEQUENTIAL SEARCH WORKS

- Start from the first element
- Compare it with the target value
- If it matches the search stops
- If not it will move to the next element
- Continue until element is found or list ends

ALGORITHM STEPS

- Take the list and the target value
- Start from index 0
- Compare each element with the target
- If found → return index
- If not found after full traversal → return -1

PYTHON PROGRAM

```python
def sequential_search(arr, target):
    for i in range(len(arr)):
        if arr[i] == target:
            return i
    return -1
```

The output will be either
the element ‘ i ‘ that we need

or
it will give -1 indicating that the element is not found

// EXAMPLE //

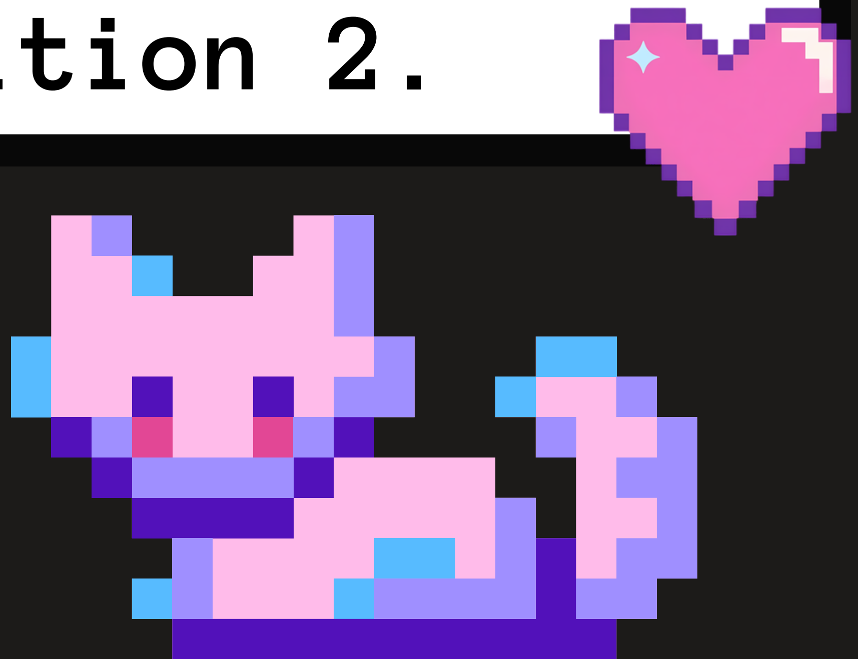
```
arr = [10, 20, 30, 40]
target = 30

result = sequential_search(arr,
target)
print(result)
```

output becomes:

2

30 is found at index position 2.



ADVANTAGES

- Very simple and easy to understand
- Works on unsorted data
- No extra memory required
- Good for small datasets
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DISADVANTAGES

- Slow for large datasets
- Requires checking every element
- Not efficient compared to binary search
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That's All Folks!